

# Agentic Workflow — Complete Reference

Last updated: 2026-08-16 · Living document, edit this directly for day-to-day changes. As of 2026-08-16 this reference is version-controlled at `docs/agentic-workflow.md` inside the `dotfiles` repo (it previously sat loose in the untracked parent folder), so it clones to a fresh machine with the rest of the setup. `agentic-workflow.pdf` is a periodic polished snapshot generated from this file's content; only worth regenerating after a batch of real changes, not every edit. Regenerate it with `docs/build-pdf.sh` (Markdown -> HTML -> headless-Chrome print-to-PDF).

**Current state:** fully built and verified working. WezTerm + herdr + Neovim + Claude Code + Codex + OpenCode, all wired to local ( `omlx` ) and cloud (Claude) inference, voice input via Parakeet v3, the full axi tool family, and Kun's productivity stack (lavish, treehouse, no-mistakes, gnhf, firstmate) are all installed, declared in Nix, and tested end-to-end. GitHub Copilot CLI ( `gc` ) was added 2026-07-11 and authenticated 2026-08-16, now verified working end-to-end, see §2. Codex's default local model swapped again 2026-07-12, to `mlx-community--Qwen3-Coder-Next-8bit` — see §5. That session also uncovered and fixed a real bug in `rebuild.sh` (missing `--impure` ) that had been silently zeroing out `OMLX_API_KEY` on every rebuild — see §9.

## 1. Quick Reference — Keys & Commands

The one mental model: a prefix key (herdr = `Ctrl+b` ) precedes multiplexer commands. In Neovim, a leader key ( `Space` ) precedes custom commands. Press prefix/leader, release, then the next key.

### herdr — prefix `Ctrl+b`

| Key                         | Action                                   |
|-----------------------------|------------------------------------------|
| <code>Ctrl+b "</code>       | split pane horizontal                    |
| <code>Ctrl+b %</code>       | split pane vertical                      |
| <code>Ctrl+b h/j/k/l</code> | move focus                               |
| <code>Ctrl+b c</code>       | new tab (fresh shell)                    |
| <code>Ctrl+b &amp;</code>   | close tab                                |
| <code>Ctrl+b w</code>       | workspace picker                         |
| <code>Ctrl+b g</code>       | go to (jump)                             |
| <code>Ctrl+b y</code>       | copy mode (v/Space select, y/Enter copy) |

Sessions persist on a background herdr server — closing the terminal doesn't kill your work. But see §9 for why that same persistence can cause stale-environment bugs.

### Neovim — leader `Space`

| Key                        | Action                               |
|----------------------------|--------------------------------------|
| <code>h j k l</code>       | left/down/up/right                   |
| <code>w / b</code>         | next/prev word                       |
| <code>0 / \$</code>        | start/end of line                    |
| <code>gg / G</code>        | top/bottom of file                   |
| <code>Nj / Nk</code>       | jump N lines (relative #s in gutter) |
| <code>i / a / o / O</code> | insert modes                         |
| <code>Esc</code>           | leave insert + saves (custom map)    |
| <code>dd / yy / p</code>   | delete / copy / paste line           |
| <code>u / Ctrl+r</code>    | undo / redo                          |

| Key     | Action                                                                  |
|---------|-------------------------------------------------------------------------|
| Space f | find files                                                              |
| Space s | search/grep project                                                     |
| Space b | switch buffers                                                          |
| Space e | file browser (Oil)                                                      |
| Space g | git UI (Neogit)                                                         |
| Space m | markdown preview, floating window (glow.nvim), <code>:q</code> to close |
| gd      | go to definition                                                        |

### Agents

| Alias | Command                                                                                                                                                                                                                                                           |
|-------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| cc    | <code>claude --dangerously-skip-permissions</code> (cloud)                                                                                                                                                                                                        |
| co    | <code>codex --sandbox workspace-write --ask-for-approval never</code> (routed to local <code>omlx</code> , default model <code>mlx-community--Qwen3-Coder-Next-8bit</code> as of 2026-07-12 — see §5)                                                             |
| oc    | <code>omlx launch opencode --model mlx-community--Qwen3.6-35B-A3B-8bit</code> (routed to local <code>omlx</code> ). Still on the older default — not yet swapped to Qwen3-Coder-Next, see §5. Prefer this over <code>co</code> for context-heavy work regardless. |
| gc    | <code>copilot --allow-all</code> (cloud, GitHub Copilot subscription). Added 2026-07-11, authenticated 2026-08-16 and working, see §2.                                                                                                                            |

`/` — slash commands (skills, config, `/lavish`, `/no-mistakes`). Plan mode — ask the agent to plan before building. `Esc` — interrupt agent mid-task.

### Voice input (Parakeet v3)

- hold `⌘ Option Right` — push-to-talk: hold, speak, release
- `⌘ Option + `` — toggle: press to start, press to stop

Global hotkey — works in any focused app (herdr pane, Messages, anywhere), agent-agnostic. No vocab-priming with Parakeet (Whisper-only feature) — see §6.

### Git essentials

| Command                    | Action                                 |
|----------------------------|----------------------------------------|
| <code>git status</code>    | what changed                           |
| <code>add</code>           | alias: <code>git add .</code>          |
| <code>git commit -m</code> | commit staged                          |
| <code>push / pull</code>   | aliases for <code>git push/pull</code> |
| <code>m</code>             | alias: <code>git switch main</code>    |
| <code>lazygit</code>       | full-screen git TUI                    |

Protocol: HTTPS via `gh auth login`, not SSH — background daemons (no-mistakes, gnhf, firstmate crew) can't reliably reach an SSH agent socket, but the gh-issued token in Keychain works for any process.

### omlx (local inference)

| Command                   | Action                          |
|---------------------------|---------------------------------|
| <code>omlx status</code>  | server/client health            |
| <code>omlx restart</code> | restart + rescan for new models |

| Command                                                  | Action                                                                                                        |
|----------------------------------------------------------|---------------------------------------------------------------------------------------------------------------|
| <code>omlx launch &lt;tool&gt; --model &lt;id&gt;</code> | wire a tool to omlx                                                                                           |
| <code>hf download &lt;repo&gt;</code>                    | pull a new model (omlx's bundled CLI, <code>/opt/homebrew/Cellar/omlx/&lt;version&gt;/libexec/bin/hf</code> ) |

## 2. Component Status

| Component                                     | Status  | Notes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|-----------------------------------------------|---------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| WezTerm, herdr, Neovim, Claude Code, node/npx | done    | core ship, unchanged since original setup                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| lavish                                        | done    | project + global skill; visual planning                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| skill-creator                                 | done    | Anthropic's own — lets agent author new skills                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| gh-axi                                        | done    | GitHub ops, ~3x cheaper/2x faster than GitHub MCP                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| chrome-devtools-axi                           | done    | browser automation, ~40% fewer tokens than raw MCP                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| quota-axi                                     | done    | usage tracking across Claude/Codex/Cursor/Copilot/Grok                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| treehouse                                     | done    | pooled git worktrees, <code>~/local/bin/treehouse</code>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| no-mistakes                                   | done    | validation pipeline, daemon running                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| gnhf                                          | done    | "good night, have fun" — the unattended overnight autonomous loop. Via <code>npx gnhf</code> (no global install — see §9)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| firstmate                                     | done    | cloned to <code>~/github/kunchenguid/firstmate</code> ; fresh-machine re-clone step now documented in the dotfiles README                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| gh (GitHub CLI)                               | done    | authenticated, HTTPS protocol                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| omlx                                          | done    | v0.5.0. GPU memory ceiling + context-window policy hardened 2026-07-10 — see §5                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Codex CLI → omlx                              | done    | wired to omlx, verified via <code>codex exec</code> . Bundled model catalog doesn't recognize local models — cosmetic metadata warning only, see §5                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| OpenCode (oc) → omlx                          | done    | wired via <code>omlx launch opencode</code> ; preferred over Codex for context-heavy sessions since omlx writes real metadata into its config — see §5                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| Claude Code → omlx                            | decided | deliberately cloud-only — not wired, by choice                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| GitHub Copilot CLI (gc)                       | done    | cask <code>copilot-cli</code> declared in <code>configuration.nix</code> , alias in <code>home.nix</code> , installed via <code>./rebuild.sh</code> 2026-07-11. Authenticated via <code>copilot login</code> and verified working end-to-end 2026-08-16. Cloud-only for now; it does support BYOK to an OpenAI-compatible endpoint via <code>COPILLOT_PROVIDER_BASE_URL</code> ( <code>copilot help providers</code> ), so wiring it to omlx like <code>co / oc</code> is possible but not yet attempted — see §5. No global instructions file (no <code>~/copilot/AGENTS.md</code> equivalent) — see §3.                                                                                                                        |
| Markdown preview (glow + glow.nvim, Space m)  | pending | <code>glow</code> added to <code>home.packages</code> in <code>home.nix</code> 2026-07-11; <code>glow.nvim</code> added as new <code>lua/plugins/markdown.lua</code> , lazy-loaded on first <code>:Glow</code> call. Declared, not yet applied - run <code>./rebuild.sh</code> to install <code>glow</code> , then launch <code>nvim</code> once so lazy.nvim fetches the plugin.                                                                                                                                                                                                                                                                                                                                                |
| leetcode.nvim (:Leet)                         | done    | <code>kawre/leetcode.nvim</code> added as <code>lua/plugins/leetcode.lua</code> 2026-07-11 — no Nix/ <code>rebuild.sh</code> involved, this repo's Neovim plugins are managed entirely by <code>lazy.nvim</code> (installs from GitHub on next <code>nvim</code> launch). <code>lang = 'python3'</code> , <code>plugins.non_standalone = true</code> so <code>:Leet</code> works inside a normal session with other buffers open — close it with <code>:Leet exit</code> , not <code>:q</code> . Sign-in needs the full <code>Cookie</code> request header ( <code>csrftoken</code> + <code>LEETCODE_SESSION</code> ) copied from DevTools Network tab, not <code>Set-Cookie</code> — see §9 gotchas for the paste/Brave quirks. |
| OpenSuperWhisper + Parakeet v3                | done    | installed, permissions granted, verified pasting text                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |

| Component                 | Status    | Notes                            |
|---------------------------|-----------|----------------------------------|
| Kokoro TTS (voice output) | not built | bundled in omlx, unused — see §8 |

### 3. Memory Files & Skills — How the Agent Learns Your Preferences

**Global memory** ( `~/claude/CLAUDE.md` , symlinked to `AGENTS.md` for Codex/OpenCode too) — personal preferences, loaded into every session. Keep it short (~27 lines); it costs tokens on every request. Yours already has: never use em dash, don't weight development cost in technical decisions, reproduce bugs end-to-end before fixing, be pixel-perfect on UI, fix lint/test issues you notice even if unrelated to the task.

**Copilot CLI is the odd one out:** as of 2026-07-11, GitHub's docs describe no user-level/global instructions file for `gc` (no `~/copilot/AGENTS.md` equivalent to symlink `home/AGENTS.md` to). It does read per-repo `.github/copilot-instructions.md` , `.github/instructions/**/*.instructions.md` , and a repo-root `AGENTS.md` (this dotfiles repo already has one) — so it inherits project conventions, just not your personal global preferences. `copilot init` can scaffold the per-repo file. Revisit if GitHub adds a global option later.

**Project memory** ( `CLAUDE.md` / `AGENTS.md` in a repo root) — grows by correcting the agent and telling it to "remember." This is the collective learning of every session in that project.

**Skills** — conditionally-loaded procedures (only read when relevant), installed via `npm skills add <owner/repo> --skill <name> [-g]` . Move verbose/conditional instructions out of memory files and into skills to keep the memory file lean.

**Skill security:** a skill can run anything on your machine — API key exfiltration, arbitrary shell access. Do not install skills from unverified sources, even popular ones (a benchmark cited in Kun's video showed a 177k-star skill making agent output measurably worse while using 5% more tokens). Only the official kunchenguid axi catalog (`axi.md`) and Anthropic's own skills were installed here. Scanner "Med Risk" ratings on `gh-axi` / `chrome-devtools-axi` / `quota-axi` reflect that they can act on GitHub/a browser/your usage data — that's the capability working as intended, not a red flag on the code itself.

### 4. The Workflow Loop

1. **Plan** — Discuss in lavish (interactive HTML artifact, browser-based annotation) — decide requirements here, not by reading walls of text
2. **Build** — Agent works the middle — step away, start other tasks
3. **Validate** — no-mistakes: branch → adversarial review → e2e test (with evidence) → docs → lint → PR, babysat until merged
4. **Judge** — Review the PR, weighted by its risk assessment — not every diff
5. **Scale** — treehouse (parallel worktrees, no manual cleanup) + firstmate (manages the crew so you talk to one agent, not many)

**Captain's mindset:** you're an engineering director, not a diff-reviewer. Set good process, trust the pipeline, spend your energy on what to build — competition, users, direction — not line-by-line review.

**Daily muscle-memory drills:** split a herdr pane, run an agent in one, nvim in another · jump with relative line numbers ( `Nj` / `Nk` ) instead of scrolling · `Space f` a file, `Space s` a symbol — no mouse, ever · dictate instead of type whenever the prompt isn't a URL/path · detach herdr, reattach — the session is still there.

Goal: hands never leave the keyboard. That's the whole point — staying in flow.

### 5. Local Inference — omlx

omlx is an MLX inference server purpose-built for coding agents on Apple Silicon: continuous batching, paged SSD KV-caching (repeated large contexts go from 30–90s time-to-first-token down to 1–3s), and native OpenAI + Anthropic API compatibility. Verified legitimate via a real discussion thread in Apple's own `ml-explore/mlx` GitHub repo.

## Current setup

- Installed via private Homebrew tap `jundot/omlx`, trusted and declared in `configuration.nix`
- Running as a background service ( `homebrew.mxcl.omlx` launchd job), auto-starts at login
- Codex's default model as of 2026-07-12:** `mlx-community--Qwen3-Coder-Next-8bit` (80B total / 3B active MoE, 256K native context). Replaced `mlx-community--Qwen3.6-35B-A3B-8bit` — see "Model swap: → Qwen3-Coder-Next-8bit" below for rationale.
- `oc /OpenCode` is still on `mlx-community--Qwen3.6-35B-A3B-8bit` — not yet swapped, since that alias hardcodes its model in `home.nix` rather than reading `~/codex/config.toml`. Revisit if Qwen3-Coder-Next proves out well through Codex.
- Codex CLI is wired to it ( `~/codex/config.toml`, provider `omlx`, base\_url `http://127.0.0.1:8000/v1` ) — verified working end-to-end via `codex exec`
- OpenCode is also wired to it via `omlx launch opencode --model <id>` (the `oc` alias) — preferred over Codex for context-heavy work, since omlx writes its real, live context-window metadata into OpenCode's config at launch time instead of Codex guessing from a static bundled catalog
- Claude Code stays cloud-only — a deliberate choice, not a gap.

**Trust lesson from this setup:** a model already on disk ( `...Claude-4.6-Opus-Deckard-Heretic-Uncensored-Thinking-8bit` ) was deleted before use. "Heretic" is a known automated ablation technique that strips safety/refusal training — it measurably degrades coherence and instruction-following even when done carefully, and there's no legitimate coding/planning task that benefits from an uncensored model. The name also falsely implied lineage with Claude Opus. Before running any downloaded model: check the name for uncensored/ablated branding and fabricated lineage claims to closed frontier models. Prefer official mlx-community releases.

## Model swap: Qwen3.6-27B-8bit → Qwen3.6-35B-A3B-8bit (2026-07-11)

Motivation: the dense 27B model was clearing ~16.8 tok/s through Codex, well under the ~30 tok/s target for a smooth interactive loop.

### What was tried and ruled out:

- `antirez/ds4` ("DwarfStar") — a dedicated inference engine for DeepSeek V4 Flash (284B total / 13B active). Legitimate, MIT-licensed, real published benchmark of 34.27 tok/s / 25.9 tok/s (short/11.7K-context) on an M5 Max 128GB — the exact machine class here. Not adopted yet: beta-quality software, a separate engine/server from omlx (not a swap-in), and `ds4-agent` is explicitly alpha. Worth revisiting if the omlx-native option below stops being enough.
- GLM-5.2 and MiniMax-M3 top the open-weight intelligence leaderboards (Artificial Analysis) but are physically too large for 128GB (753B and 427B total params respectively, even at aggressive 4-bit quant).

**What was adopted:** `mlx-community/Qwen3.6-35B-A3B-8bit` — same family omlx already has custom kernels for (Qwen3.5/3.6), MoE with only ~3B active params per token, zero new integration (just `hf download` + `omlx restart`, no new alias structure, Codex/OpenCode stay wired exactly as before).

### Measured results (own hardware, this M5 Max 128GB):

- Raw `mlx_lm.benchmark`, short prompt: **~104 tok/s** generation (vs ~16.8 tok/s baseline)
- Raw `mlx_lm.benchmark`, ~11.7K context: **~94 tok/s** generation — MoE barely degrades under long context, unlike the dense model
- Through the live omlx server ( `/v1/chat/completions` ): ~88 tok/s
- Real `codex exec` run, identical prompt on both models: 42.3s wall-clock (27B, prompt-cache warm) vs 14.3s (35B-A3B, prompt-cache cold — a harder case for it) while producing 53% *more* output tokens
- Net: ~5.6x faster generation throughput, ~2.9x faster wall-clock on a real Codex task**

### Quality tradeoff (Qwen's own published benchmarks, direct comparison of these two exact checkpoints):

| Benchmark          | Qwen3.6-27B (old default) | Qwen3.6-35B-A3B (new default) | Gap  |
|--------------------|---------------------------|-------------------------------|------|
| SWE-bench Verified | 77.2                      | 73.4                          | -3.8 |
| SWE-bench Pro      | 53.5                      | 49.5                          | -4.0 |
| Terminal-Bench 2.0 | 59.3                      | 51.5                          | -7.8 |
| LiveCodeBench v6   | 83.9                      | 80.4                          | -3.5 |
| MMLU-Pro           | 86.2                      | 85.2                          | -1.0 |
| GPQA Diamond       | 87.8                      | 86.0                          | -1.8 |

| Benchmark | Qwen3.6-27B (old default) | Qwen3.6-35B-A3B (new default) | Gap  |
|-----------|---------------------------|-------------------------------|------|
| AIME26    | 94.1                      | 92.7                          | -1.4 |

The dense model wins every benchmark, but mostly by 1-4 points; Terminal-Bench (closest proxy to real agentic CLI tool-use) shows the largest gap at ~8 points. Judged worth it for the ~3-5.6x speed gain in an interactive loop where iteration speed compounds.

**Fallback:** the old 27B weights are kept on disk (not deleted). For the rare task where you want max quality over speed:

```
omlx launch codex --model mlx-community--Qwen3.6-27B-8bit
```

Model swap: Qwen3.6-35B-A3B-8bit → Qwen3-Coder-Next-8bit (2026-07-12, Codex only)

Motivation: newer Qwen3-Next hybrid architecture, still only ~3B active params (so generation speed stays close to the 35B-A3B default) but 80B total params and 256K native context vs the old model's much smaller window — worth it given 128GB unified memory has plenty of headroom for the 8bit quant (~79GB on disk).

**Adopted:** `mlx-community/Qwen3-Coder-Next-8bit` via omlx's bundled `hf` CLI ( `/opt/homebrew/Cellar/omlx/<version>/libexec/bin/hf` download `mlx-community/Qwen3-Coder-Next-8bit` ), same quant precision (8bit) as both prior defaults for consistency.  
`~/.codex/config.toml` updated: `model = "mlx-community--Qwen3-Coder-Next-8bit"` (provider stays `omlx` ).

**Verified:** started the omlx server, confirmed the model in `GET /v1/models` , sent a direct `/v1/chat/completions` test request (loaded + responded correctly, ~12.4s cold load), then ran a real `codex exec` end-to-end (quicksort/mergesort/heapsort request) and got correct, well-formed output.

**Measured results (own hardware, this M5 Max 128GB), same methodology as the earlier swap:**

|                                                                                         | Qwen3.6-27B-8bit (dense) | Qwen3.6-35B-A3B-8bit (MoE)        | Qwen3-Coder-Next-8bit (MoE)                                     |
|-----------------------------------------------------------------------------------------|--------------------------|-----------------------------------|-----------------------------------------------------------------|
| Short-context generation                                                                | ~16.8 tok/s              | ~104 tok/s raw / ~88 tok/s server | ~79 tok/s raw ( <code>mlx_lm.benchmark -p 64 -g 256</code> )    |
| ~11.7K-context generation                                                               | not measured             | ~94 tok/s                         | ~74 tok/s raw ( <code>mlx_lm.benchmark -p 11700 -g 256</code> ) |
| Peak memory                                                                             | 27GB-class               | ~37-39GB                          | ~85-87GB                                                        |
| Real <code>codex exec</code> wall-clock (identical quicksort/mergesort/heapsort prompt) | 42.3s                    | 14.3s (cold)                      | 9.0s (clean re-run — see caveat below)                          |

Takeaways: Coder-Next is slower per-token than 35B-A3B raw (~74-79 tok/s vs ~88-104 tok/s) — expected, it's a much bigger model (80B vs 35B total) even at the same ~3B active params, so routing/attention overhead is higher. It still shares 35B-A3B's key property: barely slows down under long context (79→74 tok/s, ~6% drop) vs. how much a dense model degrades. The 9.0s `codex exec` result being the fastest of the three shouldn't be over-read as "Coder-Next is raw-faster" — it's one sample, likely favorable output length plus a warm server, not a contradiction of the raw benchmark numbers above.

**Caveat — don't benchmark back-to-back with other heavy loads:** the first `codex exec` attempt this session measured **4:12** (vs the clean 9.0s above). Cause: two `mlx_lm.benchmark` runs (each memory-mapping ~85-87GB of weights) had just run immediately before it, evicting Coder-Next's pages from the page cache and forcing a slow cold reload under memory contention when the server needed them. Not a model or config problem — an artifact of testing methodology. Let large local-model loads settle before timing the next one.

Only `co` /Codex was switched — `oc` /OpenCode's model is hardcoded in the `home.nix` alias (not read from `config.toml` ), so it's still on `Qwen3.6-35B-A3B-8bit` until deliberately changed.

**Fallback:** both older models ( `Qwen3.6-35B-A3B-8bit` , `Qwen3.6-27B-8bit` ) are kept on disk, same as always — swap the `model` line in `~/.codex/config.toml` back if needed, no rebuild required (that file isn't Nix-managed).

Codex's fabricated context-window metadata (investigated, not fixed — by design)

Codex's TUI reports "Model metadata... not found" and internally uses `model_context_window: 258400` for any locally-served model — not from omlx, which correctly reports `max_model_len: 65536` via its own `/v1/models` endpoint. Codex's bundled model catalog just doesn't recognize custom/local models and substitutes a generic, provider-agnostic fallback ( $258400 = 272000 \times 0.95$ ), confirmed identical for GPT-5.5

on an unrelated provider in an upstream GitHub issue. Confirmed again 2026-07-12 via `codex debug models`: the entire bundled catalog is only 8 entries, all OpenAI's own hosted models ( `gpt-5.5` , `gpt-5.6-sol` , etc.) — no locally-served model can ever match it, so this warning is permanent and provider-agnostic, not specific to whichever local model is currently the default.

Deliberately **not** patched via Codex's `model_context_window` config override: a confirmed upstream bug (openai/codex #16068) means manually setting that value can permanently break auto-compaction after the first context overflow. Safe mitigations instead: prefer `oc` for anything context-heavy (already accurate), and for long `co` sessions run `/compact` manually rather than trusting Codex's automatic trigger. (Almost reintroduced this override during the 2026-07-12 model swap — caught and reverted before it landed; worth re-checking `~/.codex/config.toml` doesn't have a stray `model_context_window` line after any future swap.)

### GPU memory & context-window hardening (2026-07-10)

omlx hard-crashed once under memory pressure. Root cause: macOS resets the `iogpu.wired_limit_mb` sysctl to a low default on every reboot, undercutting omlx's own internal ~122GB target, so instead of gracefully rejecting an oversized request it ran out of wired memory and hard-panicked. Fixed with two durable changes:

- `iogpu.wired_limit_mb=124928` (122GB) via a `launchd.daemons` entry in `configuration.nix` with `RunAtLoad = true` — not `system.activationScripts` , which has a known nix-darwin issue where it doesn't reliably re-run after reboot (only on `darwin-rebuild switch` )
- `max_context_window_policy: 65536` in `~/.omlx/settings.json` (under `sampling` ) — caps requests server-side so an oversized context is rejected up front instead of exhausting memory

### Adding another local model later

```
/opt/homebrew/Cellar/omlx/<version>/libexec/bin/hf download mlx-community/<repo>
omlx restart # required — server doesn't hot-reload; this triggers a rescan
```

Benchmark before switching defaults:

```
/opt/homebrew/Cellar/omlx/<version>/libexec/bin/mlx_lm.benchmark --model <repo> -p 64 -g 256 -n 3
/opt/homebrew/Cellar/omlx/<version>/libexec/bin/mlx_lm.benchmark --model <repo> -p 11700 -g 256 -n 2
```

## 6. Voice Input — OpenSuperWhisper + Parakeet v3

Local, free, open-source dictation. Installed as a cask, declared in `configuration.nix` .

| Model                            | Why                                                                                                                                                |
|----------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|
| Whisper V3<br>Large/Medium/Small | not used — slower, hallucinates during silence                                                                                                     |
| Parakeet v2                      | marginally more accurate on English only (6.05% vs 6.32% WER), no multilingual                                                                     |
| <b>Parakeet v3 — chosen</b>      | beats Whisper Large V3 on accuracy (6.32% vs 7.44% WER) at ~23x the speed, zero silence hallucination, adds 25-language support at negligible cost |

**Known limitation:** vocabulary/hotword priming (feeding it jargon like "herdr", "omlx") only works with Whisper models in this app, not Parakeet. Accepted tradeoff — fall back to typing or manual correction on jargon-heavy prompts.

## 7. Not Yet Built — Kokoro TTS (Voice Output)

omlx already bundles `mlx-audio` , which includes the Kokoro TTS model — genuinely on the machine right now, just unused. Neither Claude Code nor Codex has a built-in "read the response aloud" feature. Building it would take a small hook/script: capture an agent's final response text → POST to omlx's TTS endpoint → play the resulting audio. Candidate integration points: a Claude Code Stop hook, or a wrapper skill. Not started, flagged so it isn't forgotten.

## 8. Why This Is a Different Way of Working

Every point of friction in the loop has a purpose-built fix, and they compose:

- **Typing** → voice input at ~3x typing speed, global and agent-agnostic
- **Reading walls of text** → lavish turns a plan into an annotatable visual artifact
- **Reviewing every diff** → no-mistakes catches what a human reviewer would, with recorded evidence
- **One agent at a time** → treehouse makes parallel worktrees free
- **Managing multiple sessions** → firstmate is a single point of contact for a crew
- **Tool inefficiency** → gh-axi/chrome-devtools-axi cut token cost and latency on GitHub/browser work
- **Cost/availability** → omlx gives Codex and OpenCode a zero-marginal-cost, fully local, fully private backend, with Claude Code still available for the highest-stakes work

None of this replaces judgment — it moves the job from producing and checking individual changes to directing a process that produces and checks itself.

**A real day:** morning voice-driven planning in lavish → midday parallel scaling across treehouse worktrees → afternoon judgment proportional to risk, not diff size → overnight autonomous gnhf loop → anytime cost-free iteration via `co / oc`, reaching for `cc` when stakes call for Claude's ceiling.

## 9. Maintenance, Upgrading & Future-Proofing

**Golden rule:** everything durable is declared in `~/github/JalenMickey/dotfiles/configuration.nix` (Homebrew taps/brews/casks) or `home.nix` (nixpkgs packages, env vars, session PATH). `homebrew.onActivation.cleanup = "zap"` is intentional and will silently remove anything installed ad-hoc on the next `./rebuild.sh`. Declare before your next rebuild or expect it gone.

**Never** run `brew install/reinstall/upgrade` at the same time as `./rebuild.sh`. Both touch the Homebrew Cellar and will lock-collide. Wait for one to finish before starting the other.

`sudo` commands (including `./rebuild.sh` itself) fail when run non-interactively — always run these yourself in a real terminal, not via an agent's background shell.

### Known gotchas (all hit and resolved across setup sessions)

| Symptom                                                                                                                                                                | Real cause                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Fix                                                                                                                                                                                                                            |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>home.nix</code> secret-file pattern ( <code>omlx-api-key.local</code> etc.) silently resolves to "" after a rebuild, even though the file exists and is readable | <code>darwin-rebuild switch --flake</code> runs Nix evaluation in <b>pure mode</b> by default; <code>builtins.pathExists / readFile</code> on any absolute path outside the flake's own store copy silently returns <code>false</code> /errors instead of reading the real file — confirmed directly with <code>nix eval --impure (true)</code> vs <code>nix eval (false)</code> against the same path                                                                                                                                                                                                                                                                                                                       | <code>rebuild.sh</code> now passes <code>--impure</code> to <code>darwin-rebuild switch</code> . Required for this secret-file pattern to ever work — re-check this flag survives if <code>rebuild.sh</code> is ever rewritten |
| New <code>home.nix</code> env var doesn't show up anywhere, even a fresh terminal or fully quitting and reopening WezTerm                                              | <code>herdr</code> is a persistent-session multiplexer ( <code>herdr</code> launches by "attach to persistent session", not a fresh one) — quitting/reopening the terminal app just reattaches the same long-lived pane process with its original environment. <code>herdr server stop</code> doesn't kill that pane either. Even <code>exec zsh -l</code> in that pane isn't enough by itself: <code>home-manager's hm-session-vars.sh</code> has its own re-source guard ( <code>__HM_SESS_VARS_SOURCED</code> ), and <code>exec</code> inherits environment variables from the replaced process — so the guard (already <code>1</code> from before the rebuild) blocks re-export of everything, including the fixed value | In the existing pane: <code>unset __HM_SESS_VARS_SOURCED</code><br><code>__HM_ZSH_SESS_VARS_SOURCED; exec zsh -l</code> .<br>Confirm with <code>echo \$OMLX_API_KEY</code> (or whatever var) before trusting it                |



| Symptom                                                                                                                                                                                                                              | Real cause                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Fix                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| App records/detects fine but silently does nothing (e.g. dictation doesn't paste)                                                                                                                                                    | macOS Accessibility/TCC grants don't apply to an already-running process                                                                                                                                                                                                                                                                                                                                                                                             | fully quit and relaunch the app after granting permission                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| omlx broken after a rebuild (bad interpreter path)                                                                                                                                                                                   | undeclared transitive dependency ( <code>python@3.11</code> ) got zap-removed as an "orphan"                                                                                                                                                                                                                                                                                                                                                                         | declare every real dependency explicitly in <code>configuration.nix</code>                                                                                                                                                                                                                                                                                                                                                                                                                          |
| global <code>npm install -g</code> fails outright                                                                                                                                                                                    | this machine's npm prefix is a read-only Nix store path                                                                                                                                                                                                                                                                                                                                                                                                              | use <code>npx &lt;pkg&gt;</code> for on-demand invocation                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| omlx hard-crashed under memory pressure                                                                                                                                                                                              | <code>iogpu.wired_limit_mb</code> resets to a low default every reboot                                                                                                                                                                                                                                                                                                                                                                                               | <code>launchd.daemons</code> <code>sysctl</code> entry + <code>max_context_window_policy: 65536</code> — see §5                                                                                                                                                                                                                                                                                                                                                                                     |
| <code>co</code> breaks with unexpected argument ' <code>--full-auto</code> '                                                                                                                                                         | Codex 0.128 removed <code>--full-auto</code>                                                                                                                                                                                                                                                                                                                                                                                                                         | use <code>--sandbox workspace-write --ask-for-approval never</code> — not <code>--dangerously-bypass-approvals-and-sandbox</code> , which removes the sandbox entirely                                                                                                                                                                                                                                                                                                                              |
| Codex reports fabricated context-window metadata (258400)                                                                                                                                                                            | bundled catalog doesn't recognize local models, falls back to a generic constant                                                                                                                                                                                                                                                                                                                                                                                     | no safe config fix (upstream #16068); use <code>oc</code> for context-heavy work, <code>/compact</code> manually in long <code>co</code> sessions                                                                                                                                                                                                                                                                                                                                                   |
| <code>leetcode.nvim</code> errors "contains listed buffers" on <code>:Leet</code>                                                                                                                                                    | plugin refuses to launch in a session with other files already open, by design                                                                                                                                                                                                                                                                                                                                                                                       | fixed here via <code>plugins.non_standalone = true</code> in <code>leetcode.lua</code> — close with <code>:Leet exit</code>                                                                                                                                                                                                                                                                                                                                                                         |
| <code>leetcode.nvim</code> sign-in: pasting the cookie does nothing                                                                                                                                                                  | Ctrl+Shift+V / Shift+Insert / <code>"*p</code> don't reliably paste into the prompt in some terminals                                                                                                                                                                                                                                                                                                                                                                | try <code>&lt;C-r&gt;*</code> (insert register), or your terminal's native paste; on Brave, strip the leading <code>cf_clearance=...</code> ; so the string starts at <code>csrftoken=</code>                                                                                                                                                                                                                                                                                                       |
| <code>node</code> / <code>npx</code> abort with Library not loaded: <code>.../libada.3.dylib</code> (breaks <code>npx gnhf</code> , and <code>opencode</code> / <code>oc</code> which is a Homebrew node dependent)                  | Homebrew <code>node</code> links <code>libada.3</code> , but its <code>ada-url</code> dependency auto-upgraded to 4.0.0 (only <code>libada.4</code> exists) and node wasn't relinked - Homebrew <code>onActivation.autoUpdate = true</code> is the trigger                                                                                                                                                                                                           | <code>brew reinstall node</code> pulls a current node bottle linked to <code>libada.4</code> . Do NOT <code>brew uninstall node</code> : <code>opencode</code> depends on it. Note this node is undeclared cruft - you declare <code>nixpkgs nodejs_22</code> (works at v22, shadowed on PATH), so a rebuild would zap it anyway                                                                                                                                                                    |
| <code>brew reinstall/install</code> aborts with Error: unknown install step: remove                                                                                                                                                  | nix-homebrew's patched brew (6.0.1, <code>zhaofengli/nix-homebrew</code> ) is behind homebrew-core; the current formula uses a post-install step the patched brew doesn't implement                                                                                                                                                                                                                                                                                  | the bottle itself usually still pours and links (node came out usable), so verify with <code>node -v</code> before panicking. Durable fix: <code>nix flake update nix-homebrew &amp;&amp; ./rebuild.sh</code> - never mid-brew                                                                                                                                                                                                                                                                      |
| <code>gnhf</code> (or any <code>oc</code> / OpenCode agentic run) burns iterations with "OpenCode produced no final answer", 0 commits, then bails after 3 strikes - even though the omlx server is up and answers <code>curl</code> | the model was pointed at a <b>reasoning</b> model ( <code>Qwen3.6-35B-A3B</code> ), which spends its whole output budget in a <code>&lt;think&gt;</code> stream and never lands a clean tool call. Proof: <code>curl</code> the model with "reply exactly OK" and it returns "Here's a thinking process:\n\n1." with <code>finish_reason: length</code> instead of <code>OK / stop</code> . Reasoning models are fine for chat, wrong for agentic tool-calling loops | point OpenCode at a <b>codex</b> model instead - <code>~/config/opencode/opencode.json</code> <code>"model": "omlx/mlx-community--Qwen3-Coder-Next-8bit"</code> (declare it under <code>provider.omlx.models</code> too). Verify: <code>curl</code> it with "reply exactly OK" - a codex model returns <code>OK / finish_reason: stop</code> . Same rule for Codex/ <code>co</code> , which already defaults to the Codex model (see §5). First call per model is a ~1min cold MLX load, not a hang |

## Update commands (verified)

| What                                       | Command                                                                                                                              |
|--------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|
| Skills (lavish, axi family, skill-creator) | <code>npx skills update -g</code>                                                                                                    |
| Codex CLI                                  | <code>codex update</code>                                                                                                            |
| OpenCode                                   | <code>brew upgrade opencode</code> (declared in <code>configuration.nix</code> , so <code>./rebuild.sh</code> also keeps it current) |
| herdr                                      | <code>herdr update</code> (or <code>herdr channel set stable preview</code> )                                                        |
| firstmate                                  | <code>/updatefirstmate</code>                                                                                                        |

| What                                                    | Command                                                                                                                                           |
|---------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|
| gnhf                                                    | nothing to do — <code>npx gnhf</code> always resolves latest                                                                                      |
| treehouse / no-mistakes                                 | re-run the original install script (idempotent)                                                                                                   |
| omlx                                                    | <code>brew upgrade jundot/omlx/omlx</code> — recompiles from source (~3 min); never alongside <code>./rebuild.sh</code>                           |
| Copilot CLI                                             | <code>copilot update</code> — the cask is tagged <code>auto_updates</code> , so <code>brew upgrade</code> won't manage it; the app updates itself |
| gh, node, everything else in <code>home.packages</code> | <code>nix flake update</code> then <code>./rebuild.sh</code>                                                                                      |

### Periodic health checks

- `gh auth status` — tokens can expire
- `copilot login` (re-run if `gc` starts prompting to authenticate — Copilot tokens can also expire)
- `omlx diagnose menubar` — built-in self-diagnostic
- Check disk usage under `~/omlx/models` and `~/cache/huggingface` — local models are 20-80GB+ each and accumulate
- Before downloading any new local model: re-run the uncensored/abliterated-branding sanity check from \$5
- Re-run `./rebuild.sh` occasionally even with no known changes — catches config drift early

### Secrets hygiene

The dotfiles repo is public. Pattern used for `OMLX_API_KEY` — reusable for any future credential:

```
home/<name>.local          # plaintext value, gitignored (home/*.local)
home.nix reads it via:
  lib.optionalString (builtins.pathExists file) (lib.strings.trim (builtins.readFile file))
```

This keeps the *shape* of the config declarative and public while the actual secret value never enters git history.

**This pattern requires** `--impure`. `builtins.pathExists / readFile` on the secret file's absolute path silently fails under Nix's default pure evaluation for flakes — see the gotchas table above. `rebuild.sh` already has this flag; don't drop it when adding the next secret.